MILESTONE 3

ProSense

Modular platform for intelligent functional training monitoring based on computer vision

Group 1
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Supervisors: António Neves · Daniel Canedo
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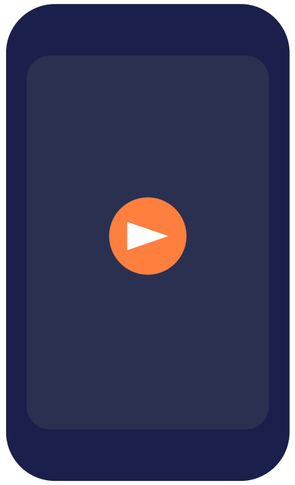




What is ProSense

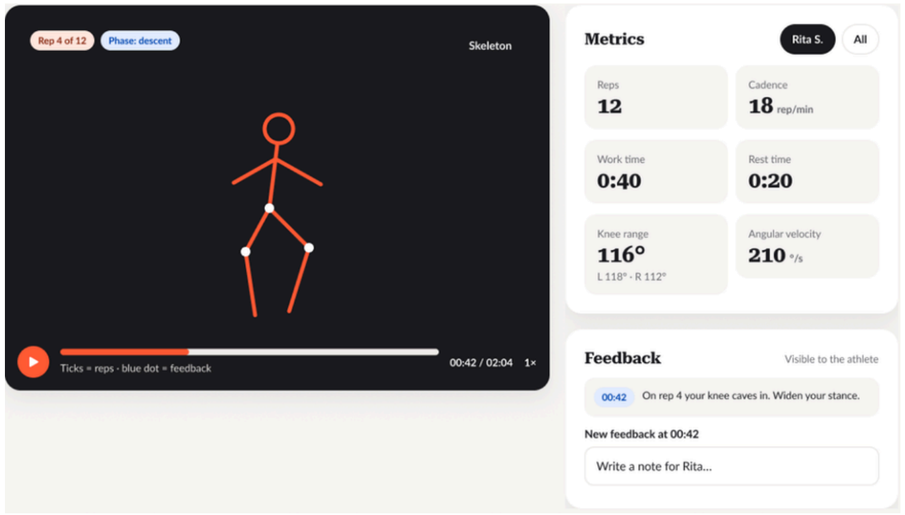
Film an exercise. Receive feedback.

Training videos become movement metrics for the coach, who gives feedback to the athlete



1 The athlete films

Uploads a training video



2 ProSense analyses, the coach reviews

Skeleton, metrics and a note at the right moment

JM

João Matos

PT · Back squat · 00:42

“On rep 4 your knees cave in. Widen your stance a little.”

See details

3 The athlete gets feedback

In the app, tied to the video

CLIENT A functional training box, partner of the project

Where we are

M1 29 Sep

Inception

Vision and scope

- ✓ Context, problem, 3 goals
- ✓ 5 user profiles
- ✓ MVP and calendar

M2 Oct

Elaboration

Requirements, architecture

- ✓ 28 use cases, 38 requirements
- ✓ Plugin-based architecture
- ✓ Mock-ups of main screens

M3 Today

Legal and accessibility

What we present today

- ✓ GDPR, licences, T&C
- ✓ 73 screens, 38 problems
- ✓ Frontend built

MVP 15-16 Dec

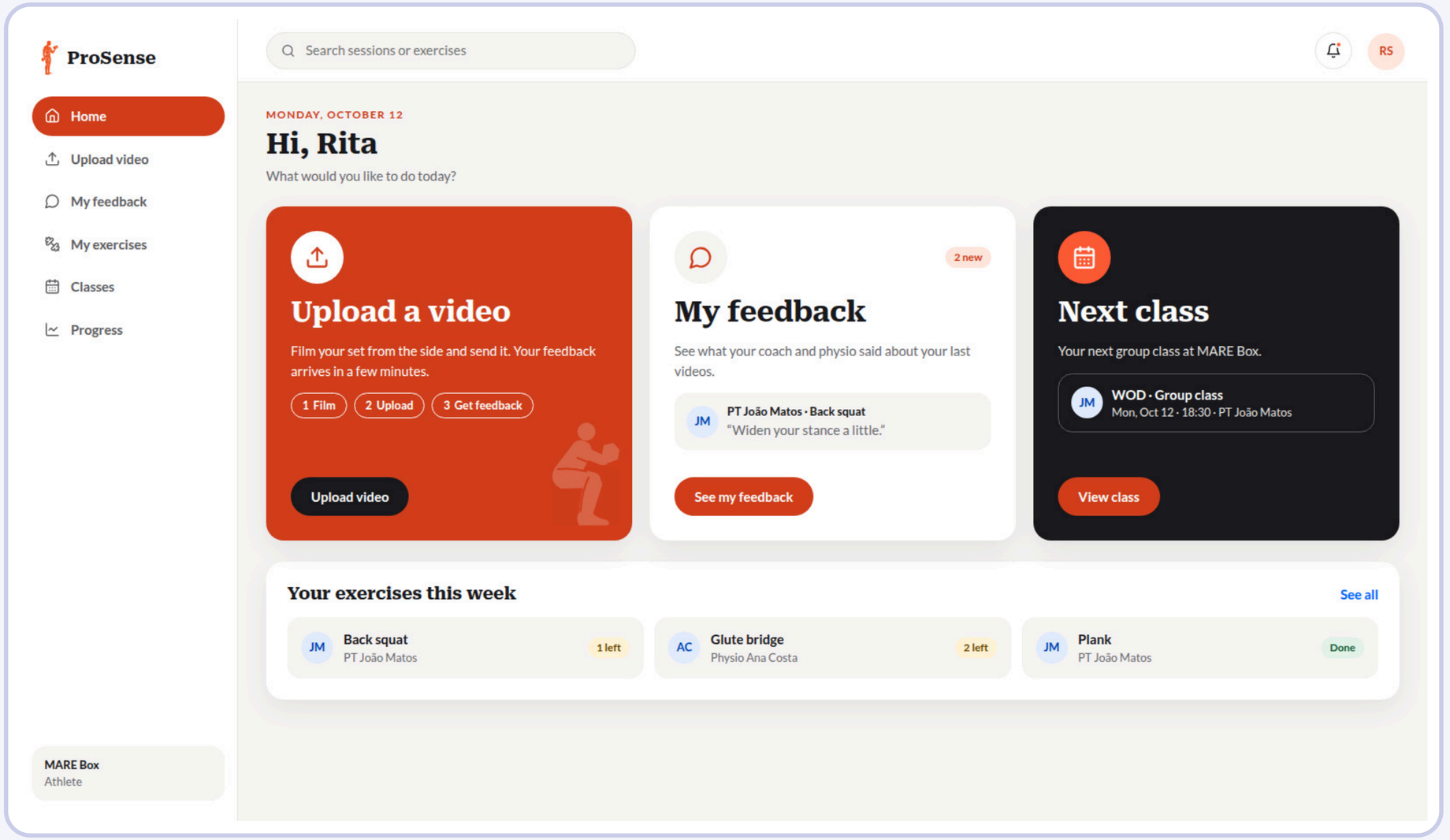
Working platform

What comes next

- ○ Upload to feedback, end to end
- ○ Two pose estimators + MDL
- ○ Real MARE Box videos

Since M2: the frontend was built and improved with the feedback we received

Changes


ProSense
Search sessions or exercises
RS
Home
Upload video
My feedback
My exercises
Classes
Progress
MONDAY, OCTOBER 12
Hi, Rita
What would you like to do today?
Upload a video
Film your set from the side and send it. Your feedback arrives in a few minutes.
1 Film
2 Upload
3 Get feedback
Upload video
2 new
My feedback
See what your coach and physio said about your last videos.
JM
PT João Matos · Back squat
"Widen your stance a little."
See my feedback
Next class
Your next group class at MARE Box.
JM
WOD · Group class
Mon, Oct 12 · 18:30 · PT João Matos
View class
Your exercises this week
See all
JM
Back squat
PT João Matos
1 left
AC
Glute bridge
Physio Ana Costa
2 left
JM
Plank
PT João Matos
Done
MARE Box
Athlete

Data protection (GDPR)

WE COLLECT

- Training videos
- Movement metrics
- Rehab data

Signed consent
rehab data = health data (Art. 9)

LIFE OF A VIDEO

Consent
Signed before recording, revocable, 18+

Upload
GPS and audio removed

Local analysis
Worker sees a code, not the name. No cloud

Day 30
Video deleted (max.), keypoints stay

Erasure
Account deleted: all data gone within 1 month

Athlete
Only their own data

Coach / Physio
Only their own athletes

Researcher
Pose data only, no images

Approved	UA DPO opinion 22-2026-EPD (ref. CEIC 6-CEIC-UA/2026-D). DPIA done, low residual risk.
Security	TLS, encryption at rest, 2FA, role-based access. Incidents: DPO notifies the CNPD in 72 h.
Scope	Everything runs through MARE Box (written authorisation, Dec 2025). Open: independent physios.

What we keep and for how long

Every item has a purpose, a legal basis and a deletion date. **Orange = health data (Art. 9)**

Identification

Name, signature; age, sex, experience

Consent · Art. 6(1)(a)

Name kept until 31 Oct 2027

Training video

Pose extraction and review by the coach

Consent · Art. 6(1)(a)

Max. 30 days

Keypoints & metrics

Results, progress, open pose dataset

Consent · Art. 6(1)(a)

Deleted 31 Oct 2029

Rehab data

Follow-up by the physio

Explicit consent · Art. 9(2)(a, h)

During follow-up (open)

Code ↔ name key

Kept apart, researcher-only access

Pseudonymisation · Art. 89

Destroyed 31 Oct 2027

Technical logs

Security and audit

Legitimate interest · Art. 6(1)(f)

90 days

Licensing and intellectual property

PERMISSIVE LICENCES: commercial use allowed, no conflict

Apache 2.0	Spring Boot · SeaweedFS · OpenCV · MediaPipe · RTMPose · Docker
MIT	React · React Native · Vite
BSD / PostgreSQL	Nginx (BSD-2) · Pika (BSD-3) · PostgreSQL
MPL 2.0	RabbitMQ

OWNERSHIP

ProSense is owned by the **University of Aveiro (IEETA)** and will **not be commercialised**. MARE Box is a partner and uses it free of charge.

THE ONLY CONFLICT

YOLO · AGPL-3.0

AGPL applies whenever others use the software over a network, even for free. ProSense source code will be public under AGPL-3.0, so YOLO can be used in the platform.

Resolved

MDL · Lúcia Sousa, IEETA

No licence defined yet: all rights reserved until IEETA decides.

Pending IEETA

FULL TABLE

All 15 components and licences are in the legal pillar document.

Medical devices and sector regulation

Regulation (EU) 2017/745 (MDR)

ProSense **supports trainers and physiotherapists**. It does not diagnose, treat or prescribe: **the professional always decides**.

Training

Coaches and athletes

- Counting reps, cadence and range of motion is fitness monitoring
- → **Not a medical device**

Rehabilitation

Physiotherapists · borderline

- Monitoring an injury is a medical purpose under the MDR
- Prescriptions and alert thresholds are set by the physiotherapist
- Low-confidence warnings, no diagnostic claims
- Research prototype, not on the market: a commercial release needs an MDR assessment

Other regulation

- GDPR: favourable UA DPO opinion (22-2026-EPD)
- AI Act (Reg. 2024/1689): not high-risk
- Law 39/2012 (gyms): the trainer prescribes, not software

Terms and conditions

Five key points that protect the software's creators

1 Terms of use

For MARE Box members aged 18+. Accounts are personal. Only record people who signed the consent form.

2 Limitation of liability

Metrics are automatic estimates, not medical advice. Exercise is done at the user's own risk.

3 Intellectual property

Belongs to the University of Aveiro (IEETA). Users keep the rights to their videos (deleted within 30 days).

4 Jurisdiction

Portuguese law (GDPR, Law 58/2019). Requests go to the responsible researcher and the UA DPO; complaints to the CNPD.

5 Changes to terms

Users are notified in advance. Continued use means acceptance.

Refunds

Not applicable: ProSense is free and there are no payments.

Digital accessibility: approach

A critical analysis of our own screens, before testing with users

73

screens analysed

login, athlete, coach, physio, researcher and admin

6

user profiles

blind, low vision, colour blindness, motor, cognitive, deaf

AA

WCAG 2.2 level

the international criteria for web and mobile

HOW WE TESTED

1. **Screen reader (NVDA)**
2. **Keyboard only**
3. **Zoom at 200%**
4. **Contrast and colour blindness**

REFERENCES

WCAG 2.2, level AA: the criteria we check

Decree-Law 83/2018 and European Accessibility Act: reference, not an obligation

Accessibility: what we found

38

problems found

High severity	4
Medium severity	20
Low severity	14

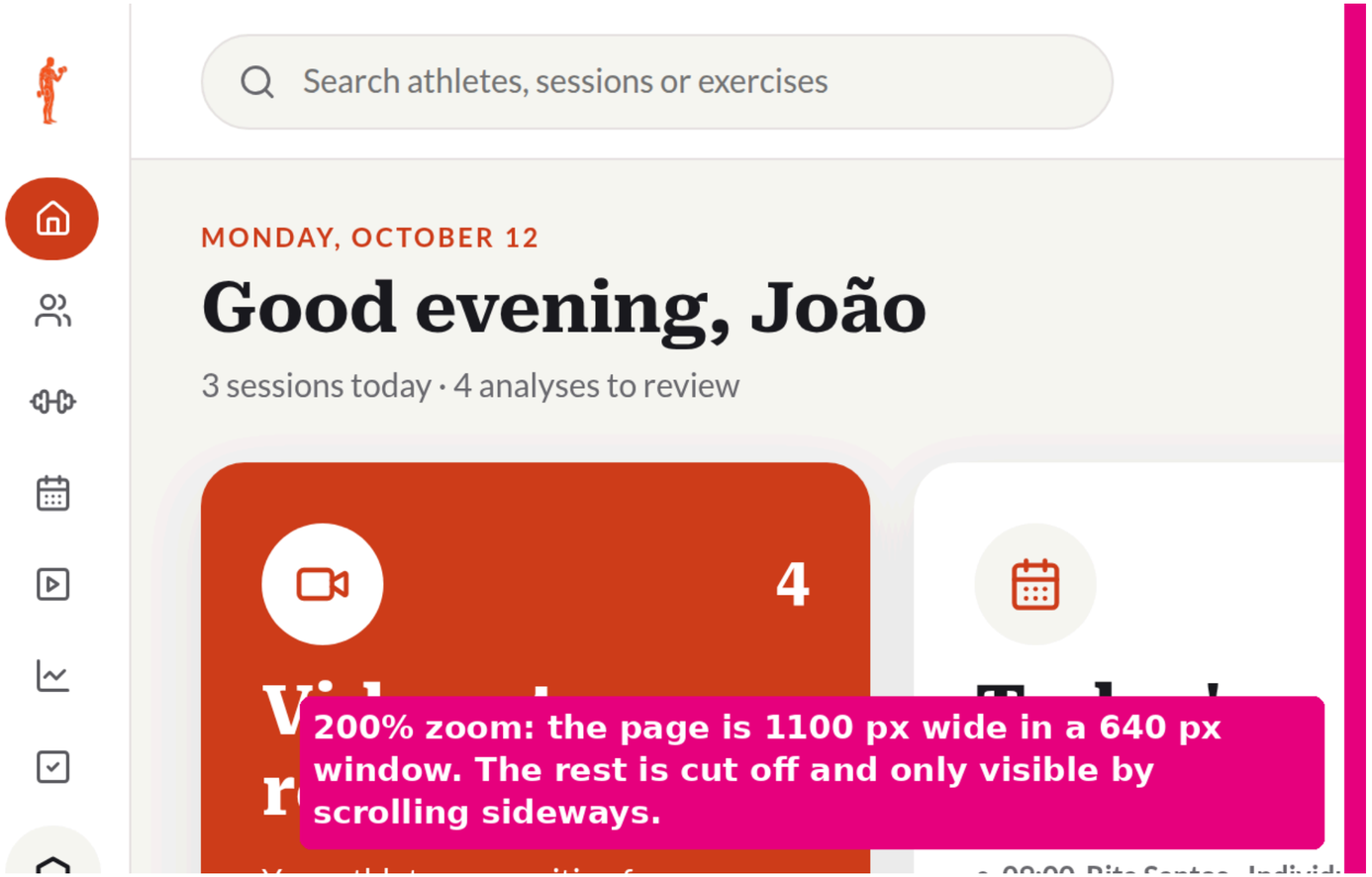
WHO IS AFFECTED

Blind users	25
Low vision	11
Motor limitations	10
Colour blindness	6
Cognitive limitations	6

Deafness: not applicable, the product has no audio

A problem can affect more than one profile.

Problem 1: Content cut off at 200% zoom



High severity

PROBLEM

Every screen keeps a fixed 1100 px width, so content is cut off and needs horizontal scrolling.

WHO IS AFFECTED

Low vision

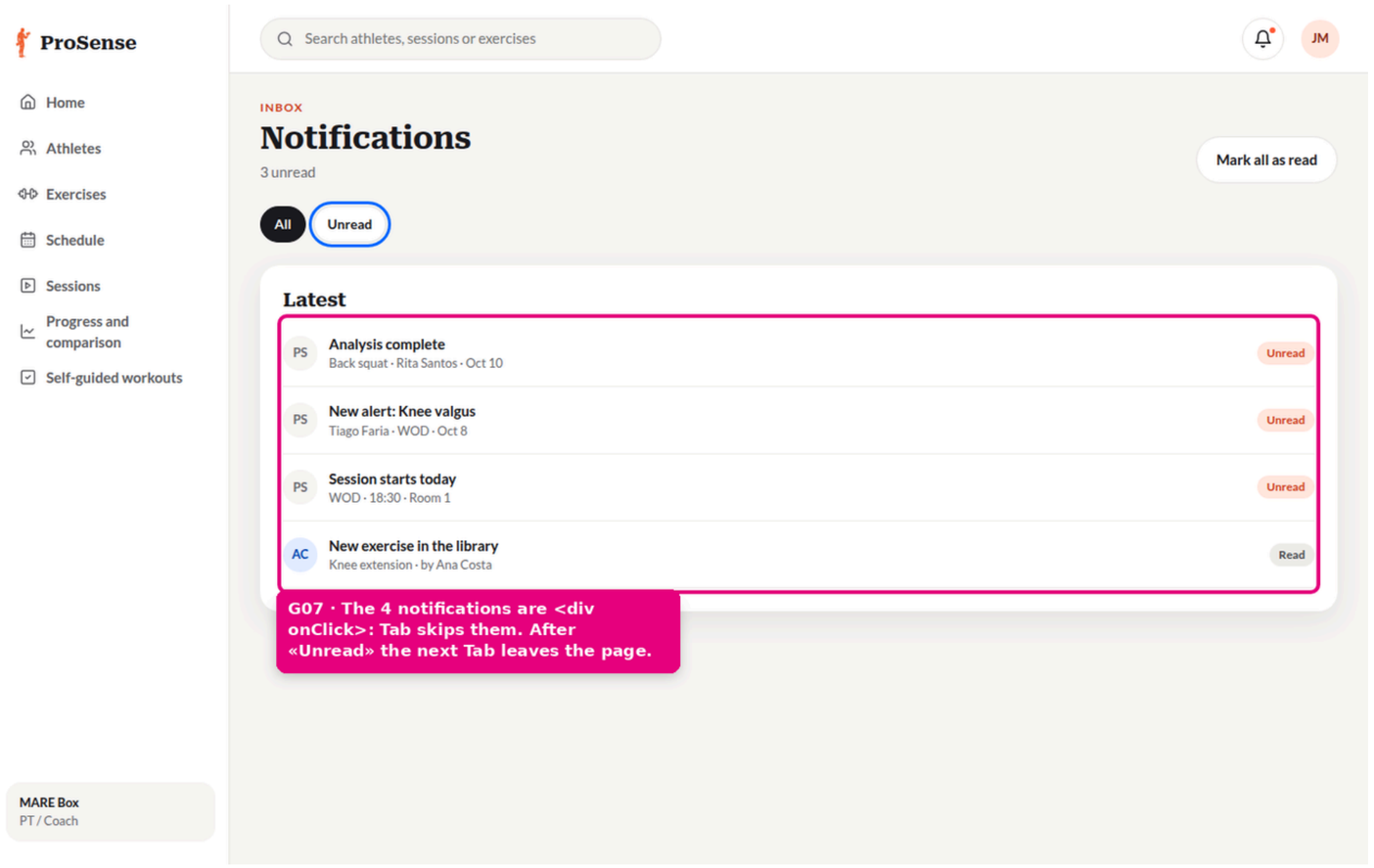
WCAG CRITERION

1.4.10 Reflow

PROPOSED FIX

Remove the fixed min-width and make the layout responsive (collapsing sidebar, stacked cards and tables).

Problem 2: Notifications need a mouse



High severity

PROBLEM

Notifications and schedule cards are clickable s that cannot be reached or opened with the keyboard.

WHO IS AFFECTED

Keyboard-only and blind users

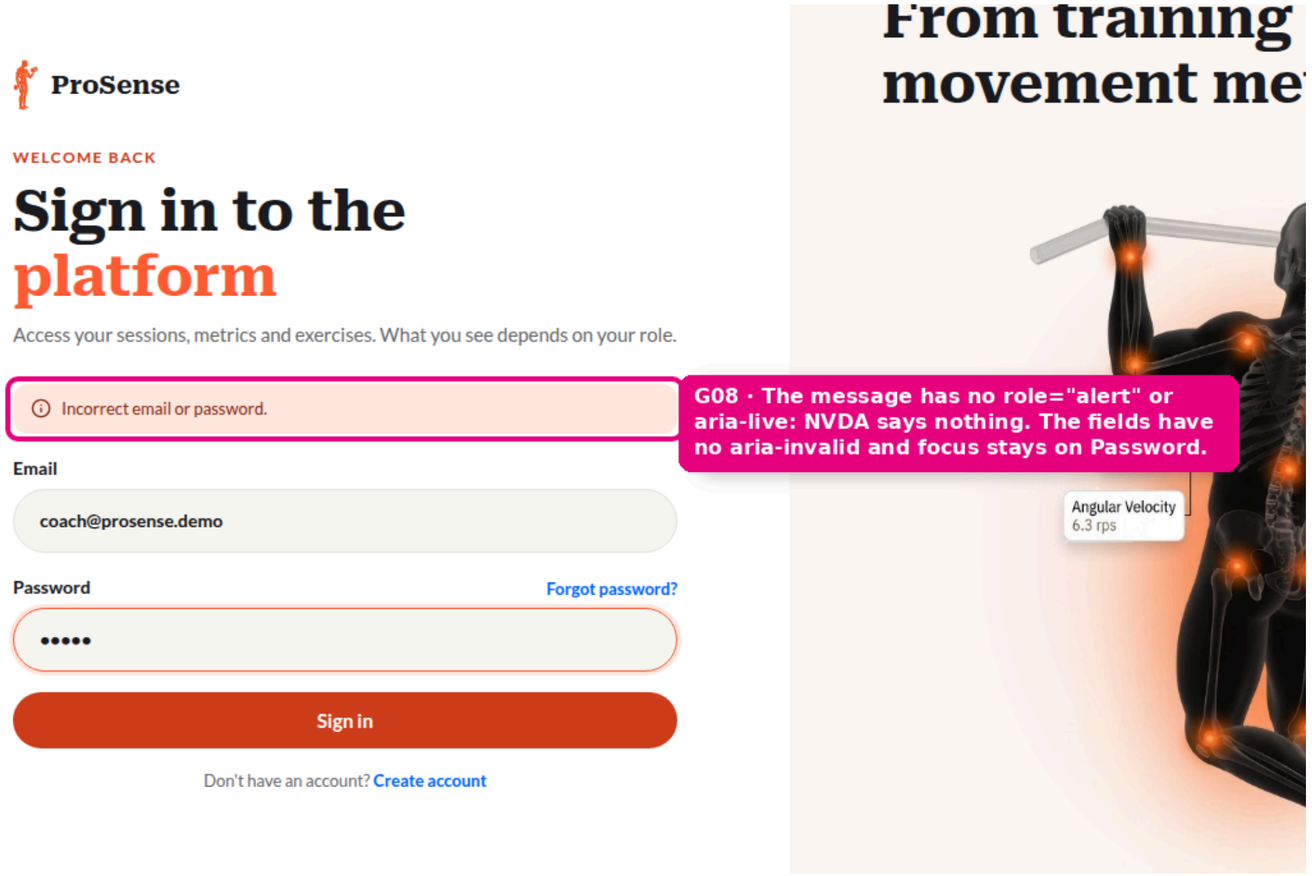
WCAG CRITERION

2.1.1 Keyboard

PROPOSED FIX

Make each item a real link or button with a visible focus style.

Problem 3: Login error not announced



High severity

PROBLEM

A failed sign-in shows an error that screen readers never announce, and focus stays in the password field.

WHO IS AFFECTED

Blind users, cognitive disabilities

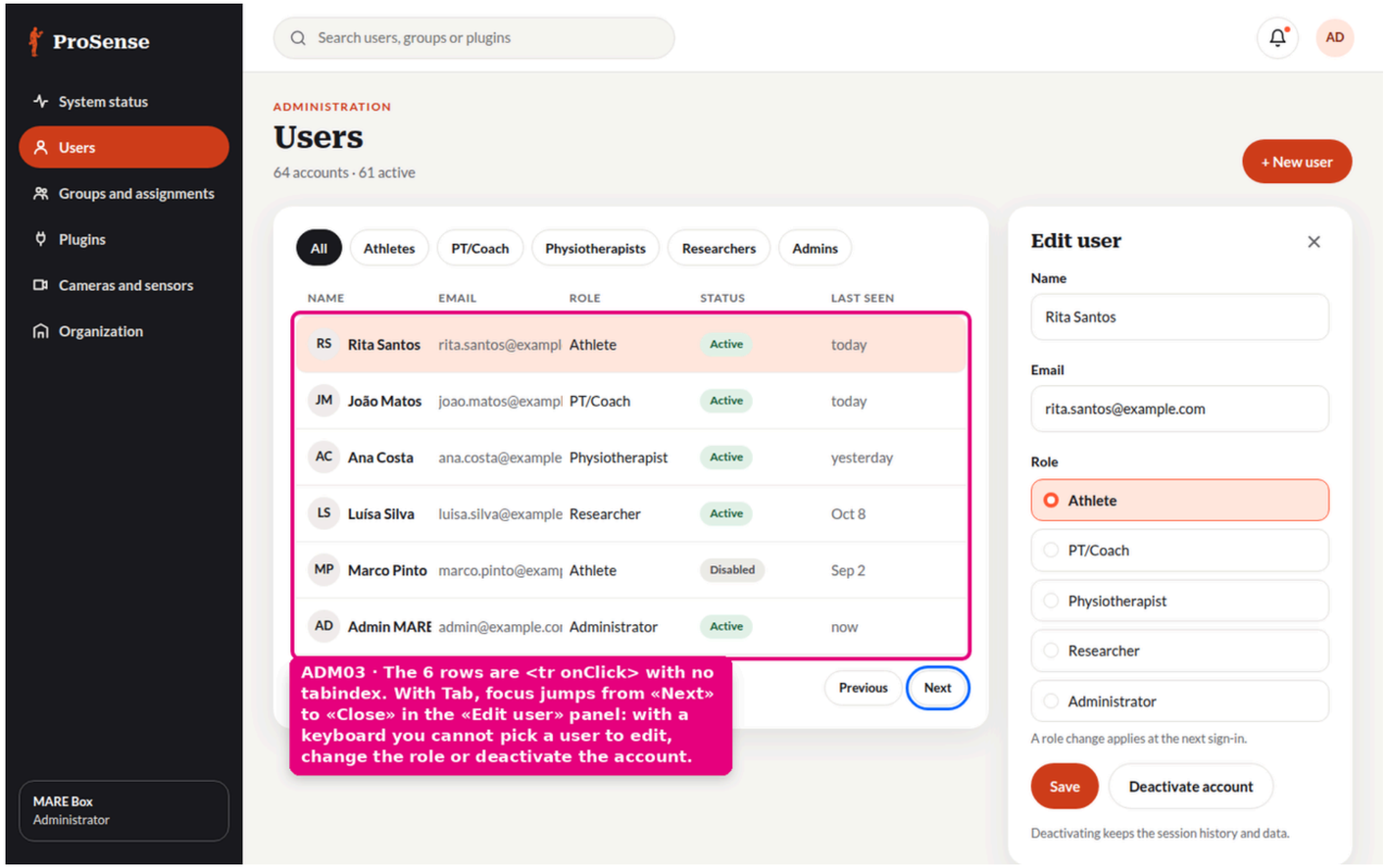
WCAG CRITERION

3.3.1 Error Identification, 4.1.3 Status Messages

PROPOSED FIX

role="alert" on the message; aria-invalid and aria-describedby on both fields.

Problem 4: Admin user list is mouse-only



High severity

PROBLEM

Table rows only react to a mouse click, so a keyboard user cannot pick a user to edit, change their role or deactivate the account.

WHO IS AFFECTED

Keyboard-only and blind users

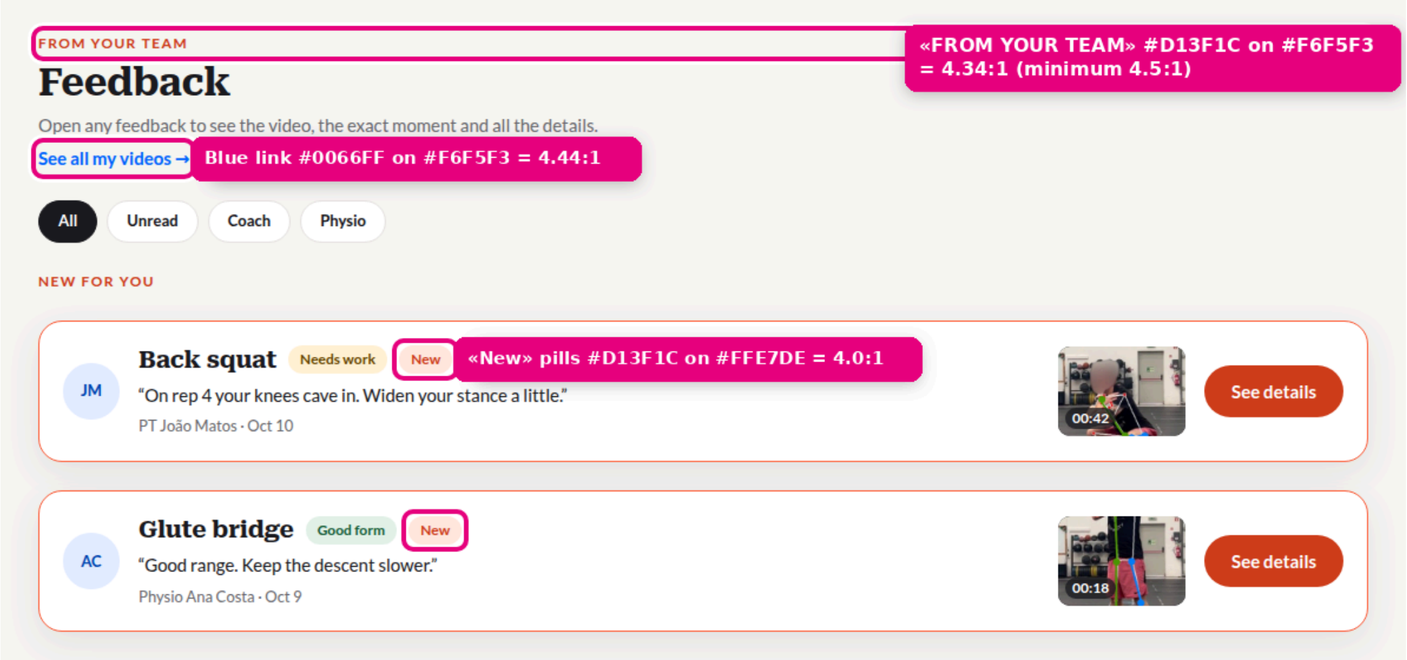
WCAG CRITERION

2.1.1 Keyboard

PROPOSED FIX

A button or link with the user's name in the first cell of each row, opening the Edit user panel.

Problem 5: Low contrast on orange text



Medium severity

PROBLEM

Small orange text (eyebrows, status pills) has 4.0 to 4.3:1 contrast and the 403/404 codes 2.85:1, below the 4.5:1 and 3:1 minimums.

WHO IS AFFECTED

Low vision

WCAG CRITERION

1.4.3 Contrast (Minimum)

PROPOSED FIX

Use the darker orange #B9330F for text (5.0:1 or more) and #0058DB for links.

Next milestone: calendar

Area	3 Nov	10 Nov	17 Nov	24 Nov	1 Dec holiday	8 Dec holiday	15 Dec
Infrastructure	Repo, CI, Docker				Deploy on VM	Docs, bug fixes	
Backend	Auth and API	Users, upload	Sessions, consent	Results, feedback	Deletion: 30 days		
Plugins	Worker contract	MediaPipe	YOLO and MDL		Real videos		
Web	Connect to API	Upload and coach screens		Dashboard	Accessibility fixes		
Delivery			Meeting 16/11	Check point 24/11		Freeze 9/12, rehearsal	MVP 15-16/12

Holidays: 1 and 8 December. MVP scope: web app, mobile as responsive web; Single User expirience

Thank you!

KEEP TRACK OF OUR WORK!



Gabriel Riquito



António Videira



David Cálix



Tiago Oliveira



Diogo Ruivo

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